

# Skills Summary

## Explaining Legal Equation Transformations

*Use this reference while completing your worksheet or when studying.*

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### Skill 1 — Naming the property behind a step

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**Rule:** A step is *legal* when the new equation has exactly the same solutions as the old one. Two families do that:

- do the same thing to **both sides** — Addition, Subtraction, Multiplication, or Division Property of Equality (the multiplier or divisor must be nonzero);
- rewrite **one side** into an equal form — Distributive Property, combining like terms.

**Example:** Solve  $2x - 5 = 11$  and justify.

**Step 1.** The  $x$  is multiplied then has 5 subtracted, so I undo in reverse order: the subtraction first, then the multiplication.

**Step 2.** Add 5 to both sides (Addition Property):  $2x = 16$ . Divide both sides by 2, nonzero (Division Property).  $\Rightarrow x = 8$

**Try it:** Solve  $5x + 4 = 24$ , naming the property on each line.

check:  $x = 4$

### Skill 2 — Deciding whether two equations match

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**Rule:** Rewriting one side is legal only when the two forms are equal for **every** value of the variable. Distribute to *all* terms inside the parentheses; a partial distribution is a different expression, so the equation it sits in is a different equation.

**Example:** Write  $3(x - 4)$  without parentheses.

**Step 1.** Nothing is being done to both sides here — I am only rewriting one side, so I need an identity, and the Distributive Property is the one that removes parentheses.

**Step 2.** Multiply the 3 by *both* terms:  $3 \cdot x$  and  $3 \cdot (-4)$ .  $\Rightarrow 3(x - 4) = 3x - 12$ , equal for every  $x$

**Try it:** Write  $5(x + 3)$  without parentheses.

check:  $5x + 15$

### Skill 3 — Finding the step that is not legal

**Rule:** Read down someone's work and ask of each line: *what was done, and was it done to both sides in full?* The three usual offenders are a change made to one side only, a partial distribution, and a division by a quantity that could be zero. Then substitute the answer into the **original** equation — a legal chain always survives that test.

**Example:** A student writes  $4x + 9 = 25 \rightarrow 4x = 34$ . Find the bad line and fix it.

**Step 1.** The 9 moved but got bigger, so the student added 9 to the right while removing it from the left — not the same operation on both sides.

**Step 2.** Subtract 9 from both sides:  $4x = 16$ , then divide both sides by 4.  $\Rightarrow x = 4$

**Try it:** A student writes  $7x - 2 = 19 \rightarrow 7x = 17$ . What went wrong, and what is  $x$ ?

check:  $x = 4$

### Skill 4 — Multiplying an inequality by a negative

**Rule:** Every equality property has an inequality twin, with one change: multiplying or dividing both sides by a **negative** number *reverses* the symbol, because negating reflects the number line and reverses order. Adding or subtracting never does.

**Example:** Solve  $-4x < 12$ .

**Step 1.** The only step needed is dividing both sides by  $-4$ , and that divisor is negative, so the symbol must turn around on that line.

**Step 2.**  $\frac{-4x}{-4} = x$  and  $\frac{12}{-4} = -3$ , with  $<$  becoming  $>$ .  $\Rightarrow x > -3$

**Try it:** Solve  $-5x > 20$ .

check:  $-4 > x$